

Detailed Design— by Shiri Guniman and Nir Lapidot

Thermal Imaging and CNN-based Machine Learning for Knee Pain Assessment.

1. Introduction:

1.1 Purpose:

This document presents the detailed design of a research-oriented project in the fields of computer vision and machine learning. The primary objective of the project is the classification of knee pain in soldiers using infrared thermal imaging and deep learning techniques. The core focus of the work lies in the development, analysis, and evaluation of a convolutional neural network (CNN) capable of identifying clinically significant knee pain based on thermal patterns.

1.2 Scope:

The project focuses on:

- Processing and analysis of thermal knee images
- Integration of visual data with clinical and demographic tabular data
- Design of a complete deep learning research pipeline
- Training and evaluation of a CNN model for binary pain classification (AKP = 1)

The scope of the document emphasizes the modeling and research aspects of the system rather than end-to-end software engineering concerns.

1.3 Overview:

This document details the data representation, image preprocessing pipeline, deep learning model architecture, training strategy, evaluation methodology, and experimental analyses conducted as part of the project.

2. Research-Oriented System Overview

The system is designed as a research pipeline for deep learning-based medical image analysis, where each stage supports the investigation of the central research question:

Can clinically significant knee pain be classified using thermal images ?

The pipeline consists of the following stages:

1. Image preprocessing and data preparation
2. Dataset construction and representation
3. CNN model training
4. Model evaluation and experimental analysis

The pipeline stages operate sequentially, where the output of each stage serves as the input to the subsequent stage.

3. Design Considerations

3.1 Assumptions

- Knee pain is associated with physiological changes that may manifest as surface temperature variations.
- Thermal images are captured under relatively controlled conditions.
- Self-reported VAS pain scores are used as reference labels for supervised learning.

3.2 Constraints

- Limited dataset size relative to typical deep learning benchmarks.
- Inter-subject variability (gender, age).
- Pain labels are subjective in nature and reflect patient-reported outcomes rather than a direct physiological measurement.

3.3 Standards

- Best practices in medical deep learning research.
- Strict separation between training, validation, and test sets.
- Prevention of data leakage across experimental splits.

4. Data Design and Representation

4.1 Image Data (Primary Modality)

The primary data modality consists of infrared thermal images of soldiers' knees, acquired during:

- The baseline visit
- The mid-trial visit
- The end-of-trial visit

Note: At each visit, performing a step-down exercise was optional.

For each subject, images are captured from four anatomically distinct regions:

- Anterior
- Medial
- Lateral
- Posterior

Dataset characteristics:

- Number of subjects: 82 soldiers.
- Total number of images: 1553.
- Number of images per subject: Each subject was intended to contribute approximately 18 images across three visits; however, the actual number varies due to missing data.
- The dataset contains thermal images with 2 different native resolutions.
- All images will be resized during preprocessing to a uniform resolution of 256 × 256 pixels.

4.2 Tabular Data (Auxiliary Modality)

The tabular dataset provides clinical context and supervision signals and includes:

- Gender (SEX)
- Knee side (LEFT / RIGHT)
- Visit index (Baseline / Follow-up / End)
- Worst pain score (VAS_WORST, 0–100)
- Average pain score (VAS_AVERAGE, 0–100)

These variables are used both for labeling and for stratified experimental analysis.

5. Deep Learning Pipeline Architecture

The project employs a full deep learning pipeline consistent with contemporary medical computer vision research:

Thermal Image + Metadata

→Preprocessing

Purpose: Prepare raw thermal knee images and corresponding tabular metadata for input into the deep learning model.

Input: Raw thermal knee images corresponding to specific anatomical knee regions, together with **tabular metadata containing patient identifiers, visit information, knee side, anatomical region, and clinical labels.**

Output: Normalized knee-region thermal images resized to a uniform input resolution and aligned with the relevant metadata and target labels.

→Dataset Construction

Purpose: Build a labeled dataset for supervised learning by associating each image with its corresponding metadata and pain label.

Input: Preprocessed images and tabular metadata.

Output: A labeled and structured dataset ready for experimental splitting.

→Train / Validation / Test Split

Purpose: Create unbiased subject-level splits for model training, hyperparameter tuning, and final evaluation while preventing data leakage.

Input: Labeled dataset (images with labels/metadata).

Output: Three non-overlapping subsets: training set, validation set, and test set.

→**CNN Training**

Purpose: Learn discriminative thermal patterns associated with knee pain.

Input: Preprocessed image tensors.

Output: Binary pain classification probability.

→**Model Evaluation**

Purpose: Quantitatively assess model performance.

Input: Trained model and test dataset.

Output: Performance metrics (Accuracy, ROC-AUC, Sensitivity, Specificity).



5.1 Image Preprocessing

- Low-resolution sample exclusion
- Thermal noise reduction
- Image intensity normalization
- Image resizing
- Data Augmentation

5.2 Feature Learning

Unlike classical approaches that rely solely on handcrafted features, this project leverages representation learning:

- Convolutional layers learn hierarchical thermal patterns
- Early layers capture local textures and gradients
- Deeper layers encode higher-level spatial structures
- Fully connected layers integrate global information for classification

5.3 Model Architecture (Conceptual)

- Stacked convolutional blocks
- ReLU activations
- Pooling layers for spatial downsampling

- Fully connected layers
- Sigmoid output layer for binary classification

6. **Training Strategy**

- Division of data into training, validation, and test subsets
- **Handling of imbalanced classes using appropriate strategies, such as class**
- weighting or balanced sampling
- Optimization using Binary Cross-Entropy loss
- Gradient-based optimization
- Monitoring of convergence and overfitting

6.1 **Functional Skeleton**

`load_image(image_path)`

Loads a thermal knee image from disk.

`preprocess_image(image)`

Applies noise reduction, normalization, resizing, and segmentation.

`construct_dataset(images, metadata)`

Associates images with labels and constructs the dataset.

`split_dataset(dataset)`

Splits the dataset into training, validation, and test subsets.

`train_cnn_model(train_set, validation_set)`

Trains the CNN model and monitors convergence.

`evaluate_model(model, test_set)`

Computes performance metrics on unseen data.

`predict_pain(image)`

Generates a pain probability prediction for a single input image.

7. **Evaluation Methodology**

Model performance is evaluated with an emphasis on clinical relevance:

- Classification accuracy
- ROC-AUC
- Sensitivity and specificity
- Analysis of false negatives due to their clinical implications
- Comparative analysis of pre- versus post-exercise conditions

8. Testing Strategy

Testing includes validation of preprocessing outputs, verification of dataset integrity, and consistency checks on model predictions.

Unit-level checks are applied to preprocessing steps, while dataset splits are verified to prevent data leakage.

9. User Interface Design

Inputs:-

~~-Thermal knee image uploaded by the user.~~

Actions:-

~~-Image upload.~~

~~-Execution of the trained CNN model for inference.~~

Outputs:-

~~-Binary classification result indicating presence or absence of significant knee pain.~~

~~-Estimated probability of clinically significant pain.~~

10. Error Handling and Logging

Potential errors are handled at different stages of the pipeline, including invalid image formats, missing metadata, and failed model inference.

Basic logging mechanisms record preprocessing failures and prediction errors to support debugging and reliability analysis.

11. Security Design

The system operates exclusively on anonymized data.

No personally identifiable information is stored or processed.

All analyses are performed locally, and no data is transmitted to external servers.

12. Error Analysis and Reliability

- Identification of misclassified cases
- Assessment of prediction consistency
- Discussion of methodological limitations

13. Performance Considerations

- ~~—Inference time per single image: typically under 30 seconds, in exceptional cases, up to 2 minutes.~~
- ~~—Model training performed offline~~
- ~~—Execution on local computational resources~~

14. Deployment Considerations

The system is designed for local execution in a research environment.

Model training is performed offline.

No cloud-based deployment or external services are used in the current implementation.

15. Maintenance and Future Research

- Expansion of the dataset
- Exploration of more advanced CNN architectures
- Integration of additional clinical variables